Computing Rate–Distortion–Perception Function for Distributed Coding with Decoder Side Information

Tianxiao Shi, Huihui Wu, Tao Guo, Yinfei Xu, and Qiaoqiao Zhou

***Abstract*—We compute the rate–distortion–perception (RDP) function for distributed function coding with decoder-only side information, where perception is measured by the expected total variation between the empirical reconstruction distribution and the target marginal. For conditionally independent observations and a binary target under Hamming distortion, we derive Blahut–Arimoto-type local-channel updates with nonincreasing objective values. Starting from the resulting local-channel solutions, posterior sorting traces the decoder frontier for fixed encoders. Because the perception loss of a mixture is determined by its aggregate reconstruction marginal, we then use a linear program that retains and mixes the reconstruction marginals of component strategies rather than their individual perception losses, while its dual guides further channel search. Together, these steps yield a computable sum-rate boundary of the RDP inner bound. This boundary and the conditional cooperative outer bound bracket the operational sum rate in binary XOR and AND examples.**

***Index Terms*—rate–distortion–perception, distributed source coding, function reconstruction, Blahut–Arimoto algorithm**

I. Introduction

Classical rate–distortion (RD) theory constrains average reconstruction error, but low distortion does not ensure the desired output distribution. Rate–distortion–perception (RDP) theory addresses this distinction by constraining both distortion and source–reconstruction distribution mismatch [1]. In distributed function coding, the issue is compounded because the reconstruction is formed from separately encoded observations.

Here encoder i observes only X_i, and the decoder combines all messages with side information Y to reproduce $T = f(\mathbf{X}, Y)$. Hence the reconstruction channel must be induced by local encoder channels and a common decoder. Side information changes the relevant posteriors and rates [2], [3], but the encoders cannot use Y or the other observations. Point-to-point RDP optimization therefore does not directly characterize the operational sum-rate function $R^\star(D, P)$.

Berger–Tung product auxiliaries provide an RD inner bound for distributed coding [4], [5], while function reconstruction can benefit from coding structures different from those used to reproduce the observations [6]. Existing multiterminal RDP studies address different aspects of the problem: Gray–Wyner RDP uses a joint encoder and separate decoders [7], whereas perception-aware distributed speech coding evaluates learned systems [8]. Neither directly evaluates the single-letter RDP inner bound for the separate-encoder function-reconstruction model considered here.

Computationally, the Blahut–Arimoto (BA) algorithm optimizes a single test channel in discrete RD problems [9]. Point-to-point RDP methods include Sinkhorn-type iterations, alternating minimization, and primal–dual optimization [10]–[12]. BA-type distributed RD methods have also been developed for logarithmic-loss models [13] and, by Tang and Yang, for indirect reconstruction with decoder side information [14]. The latter is the RD setting closest to ours and already accounts for local encoding. Extending this computation to RDP additionally requires controlling the reconstruction marginal jointly induced by the local channels and decoder, including its behavior when strategies are mixed.

We evaluate a product-auxiliary inner bound using the expected total variation between the target marginal and the empirical reconstruction distribution. Empirical and full-block perception constraints have different operational meanings and roles for randomness [15]–[18]. Under our criterion, predetermined subblocks make the empirical distribution converge to the mixture of their marginals without common randomness. Conditional independence, $p_{\mathbf{X}|Y} = \prod_i p_{X_i|Y}$, simplifies the rate constraints to $R_i \geq I(X_i; W_i \mid Y, Q)$, where Q specifies time sharing. This simplification alone does not establish inner–outer coincidence, as a supplementary three-encoder counterexample shows for the stated RD bounds in [14] and their RDP counterparts. Within the product class, distortion and the reconstruction marginal remain jointly determined by the local channels and decoder, leaving a nonconvex optimization problem.

For a binary target under Hamming distortion, we address these couplings in three stages. First, BA-type local-channel updates preserve the product structure and give nonincreasing objective values for fixed finite alphabets and multipliers, without guaranteeing global optimality. Second, with the encoders fixed, posterior sorting minimizes distortion at each prescribed reconstruction marginal, completing the decoder frontier before strategies are mixed. Third, mixing must retain the marginal information: if $a = \mathbb{P}\{T = 1\}$ and $b_j = \mathbb{P}\{\hat{T} = 1\}$ under strategy j, normalized nonnegative weights η_j give the limiting empirical perception loss $|\sum_j \eta_j b_j - a|$, rather than $\sum_j \eta_j |b_j - a|$. Opposite marginal discrepancies can therefore cancel. We retain (R_j, D_j, b_j) and optimize the time-sharing weights by a marginal-aware linear program (LP). Its

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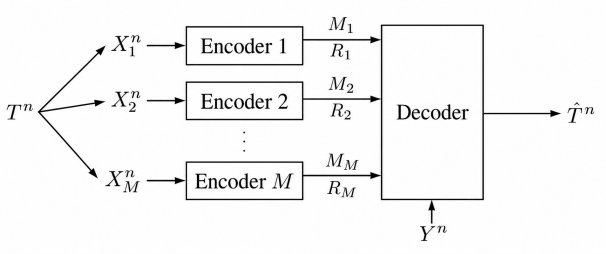


Fig. 1. Distributed coding of M observations, with Y^n available only at the decoder. Link i carries message M_i at rate R_i. The arrows from T^n do not imply independence given T.

dual supplies signed marginal multipliers for further channel searches, linking component generation, decoder optimization, and mixture refinement.

To assess the resulting finite-pool inner bound, we derive a conditional cooperative outer bound. Together, their sum-rate boundaries bracket $R^\star(D,P)$. We evaluate this bracket for XOR targets independent of Y and an AND target correlated with Y, examining how perception relaxation and target–side-information dependence affect the achievable tradeoffs.

The main contributions are summarized as follows:

- We establish a single-letter RDP inner bound and simplify its rate constraints under conditional independence.
- We combine BA-type updates, posterior sorting, and marginal-aware LP mixing to compute the sum-rate boundary of a finite-pool RDP inner bound.
- We derive a conditional cooperative outer bound and numerically bracket the operational sum rate for XOR and AND targets.

Complete proofs, implementation details, and additional results are available in the extended version [19].

II. Model and Inner Bound

Note that all alphabets throughout this paper are finite, and information quantities are in bits. Let M and n denote the encoder count and blocklength, respectively, with $[M] = \{1,\dots,M\}$ and $\mathbf{X} = (X_1,\dots,X_M)$. For $S \subseteq [M]$, write $X_S = (X_i)_{i\in S}$, $S^c = [M]\setminus S$, and $X_{-i} = X_{[M]\setminus\{i\}}$. The tuples $(T_k, \mathbf{X}_k, Y_k)$ are independent and identically distributed (i.i.d.) with probability mass function (pmf) $p_{T\mathbf{X}Y}$ supported on $T = f(\mathbf{X},Y)$ for a prescribed deterministic function f. As shown in Fig. 1, encoder i sends $M_i = e_i(X_i^n)$ with $|\mathcal{M}_i| \le 2^{nR_i}$. Only the decoder observes Y^n and forms $\hat{T}^n$ from the messages and Y^n using independent private randomness but no common randomness.

For bounded distortion d, define the empirical pmf $\hat{p}_{\hat{T}^n}(t) = n^{-1}\sum_k \mathbf{1}\{\hat{T}_k = t\}$ and

$$D_n = \frac{1}{n}\sum_k \mathbb{E}[d(T_k,\hat{T}_k)], \qquad P_n = \mathbb{E}[d_{\mathrm{TV}}(p_T, \hat{p}_{\hat{T}^n})], \quad (1)$$

where $d_{\mathrm{TV}}(p,q) = \frac{1}{2}\sum_t |p(t)-q(t)|$. A rate vector is achievable if, for every $\epsilon > 0$, codes with rates at most $R_i+\epsilon$ satisfy $\limsup D_n \le D$ and $\limsup P_n \le P$. Denote the infimum achievable sum rate by $R^\star(D,P)$. The perception criterion constrains empirical statistics, not the full block law.

Theorem 1. *Let finite Q be independent of $(T,\mathbf{X},Y)$, and let $\mathbf{W} = (W_1,\dots,W_M)$ consist of finite auxiliaries. Under local kernels q_i and decoder kernel q_d, consider the product law*

$$p_{QT\mathbf{X}Y\mathbf{W}\hat{T}} = p_Q p_{T\mathbf{X}Y} \prod_{i=1}^{M} q_i(w_i \mid x_i, q)\, q_d(\hat{t} \mid \mathbf{w}, y, q). \quad (2)$$

The closure of the union of nonnegative rate vectors satisfying

$$\sum_{i\in S} R_i \ge I(X_S; W_S \mid W_{S^c}, Y, Q), \quad \emptyset \ne S \subseteq [M],$$
$$\mathbb{E}[d(T,\hat{T})] \le D, \qquad d_{\mathrm{TV}}(p_T, p_{\hat{T}}) \le P \quad (3)$$

defines an inner bound $\mathcal{R}_{\mathrm{in}}(D,P)$.

Proof. For each $Q = q$, Berger–Tung coding [4], [5] and conditional typicality yield convergence of distortion and the empirical reconstruction pmf. Concatenating predetermined subblocks in proportions $p_Q(q)$ makes the full-block empirical pmf converge to $\sum_q p_Q(q) p_{\hat{T}|Q=q}$. Boundedness gives convergence of expectations, and closure removes rate slack. The detailed construction is given in [19]. □

When $p_{\mathbf{X}|Y} = \prod_i p_{X_i|Y}$, each subset information in (3) is additive, and the rates reduce to $R_i \ge I(X_i; W_i \mid Y, Q)$. This simplification underlies the computation below.

The product law in Theorem 1 is required by separate encoding. The RD outer bound in [14] instead imposes only $W_i - X_i - (X_{-i}, Y)$ for each i. Conditional independence given Y does not make these two auxiliary classes equivalent and therefore does not guarantee coincidence of the corresponding RDP inner and outer bounds. We establish the separation in two steps: the following construction first separates the corresponding RD inner and outer bounds at zero distortion, and exact reconstruction then makes the empirical perception loss vanish, yielding the RDP separation at $(D,P) = (0,0)$. For three independent fair bits, constant Y, and $T = X_1 \oplus X_2 \oplus X_3$, take an independent fair bit Z and

$$W_1 = 0, \qquad W_2 = Z, \qquad W_3 = Z \oplus T. \quad (4)$$

Each individual Markov condition holds, yet the auxiliaries jointly reveal T. The outer subset constraints are $\sum_{i\in S} R_i \ge I(\mathbf{X}; W_S \mid W_{S^c}, Y)$. Their right-hand sides equal zero for $S = \{1\}$ and one for every other nonempty S, admitting $(0,1,1)$. Any lossless local code requires $R_i \ge 1$, by revealing the other source blocks and applying Fano's inequality, while product auxiliaries $W_i = X_i$ attain sum rate three. Exact reconstruction gives $\hat{T}^n = T^n$ and hence $P_n \to 0$, so the same strict separation holds between the corresponding RDP inner and outer bounds at $(D,P) = (0,0)$. Full proof details and the failure of the product-replacement step are given in [19]. This separation is why the computation below remains within the locally implementable product class.

III. A Cooperative Outer Bound

Giving all observations and the side information to one encoder removes the separate-encoding restriction. Define the conditional single-terminal RDP function

$$R_{T|Y}(D,P) = \min_{\substack{q(\hat{t}|t,y):\, \mathbb{E}d(T,\hat{T})\le D \\ d_{\mathrm{TV}}(p_T,p_{\hat{T}})\le P}} I(T;\hat{T} \mid Y). \tag{5}$$

Perception remains imposed on the unconditional reconstruction marginal, as in (1). Every achievable code satisfies $R^\star(D,P) \ge R_{T|Y}(D,P)$. Indeed, with $\boldsymbol{M} = (M_1,\dots,M_M)$, message counting and data processing give

$$\begin{aligned} n\sum_i (R_i+\epsilon) &\ge H(\boldsymbol{M}) \ge I(\mathbf{X}^n;\boldsymbol{M} \mid Y^n) \\ &\ge I(T^n;\boldsymbol{M} \mid Y^n) \ge I(T^n;\hat{T}^n \mid Y^n) \\ &\ge \sum_k I(T_k;\hat{T}_k \mid Y_k). \end{aligned} \tag{6}$$

The last inequality uses the i.i.d. pairs (T_k,Y_k). A uniform coordinate index induces a conditional channel with distortion D_n and marginal loss at most P_n by Jensen's inequality. Taking limits proves the claim. When $T \perp Y$, (5) reduces to the unconditional function $R_T(D,P)$ of [1]. The complete argument and binary evaluations appear in [19].

IV. Computing the Sum-Rate Boundary of the RDP Inner Bound

Assume conditional independence given Y, binary T, and Hamming distortion, and set $a = \mathbb{P}\{T=1\}$. A component strategy $\boldsymbol{q} = (q_1,\dots,q_M,q_d)$ fixes Q and has

$$R = \sum_i I(X_i;W_i \mid Y), \quad D = \mathbb{P}\{T \ne \hat{T}\}, \quad b = \mathbb{P}\{\hat{T}=1\}. \tag{7}$$

With normalized nonnegative mixture weights η_j, perception is $|\sum_j \eta_j b_j - a|$, not $\sum_j \eta_j |b_j - a|$. We therefore store (R,D,b) for each component.

A. Local Channels and Decoder Frontiers

Use the signed-marginal objective

$$L_{\lambda,\nu}(\boldsymbol{q}) = R + \lambda D + \nu(b-a), \qquad \lambda \ge 0, \quad \nu \in \mathbb{R}. \tag{8}$$

Multipliers $\alpha,\beta \ge 0$ on $b-a \le P$ and $a-b \le P$ give $\nu = \alpha - \beta$. With all other kernels fixed, let q_i^{old} denote the current strictly positive local channel and set

$$\begin{aligned} \ell_i(x,w) &= \mathbb{E}[\lambda d(T,\hat{T}) + \nu\hat{T} \mid X_i = x, W_i = w], \\ v_i(w \mid y) &= \sum_x p_{X_i|Y}(x \mid y) q_i^{\mathrm{old}}(w \mid x). \end{aligned} \tag{9}$$

This conditioning removes the trial factor q_i, leaves ℓ_i fixed during the update, and omits zero-probability source letters. If $p_{W_i|Y}^{q_i}$ is the trial channel's induced marginal, the entropy identity

$$\begin{aligned} &\sum_{x,y,w} p_{X_iY}(x,y) q_i(w \mid x) \log_2 \frac{q_i(w \mid x)}{v_i(w \mid y)} \\ &\quad = I_{q_i}(X_i;W_i \mid Y) + \mathbb{E}_Y \, \mathrm{D}_{\mathrm{KL}}(p_{W_i|Y}^{q_i} \| v_i) \end{aligned} \tag{10}$$

gives a tight upper bound at q_i^{old}. Here I_{q_i} uses the trial channel and D_{KL} is Kullback–Leibler divergence. Adding the cost and minimizing each row gives

$$\begin{aligned} \psi_i(x,w) &= \sum_y p_{Y|X_i}(y \mid x) \log_2 v_i(w \mid y) - \ell_i(x,w), \\ q_i^+(w \mid x) &= \frac{2^{\psi_i(x,w)}}{\sum_{w'} 2^{\psi_i(x,w')}}. \end{aligned} \tag{11}$$

Setting $\nu = 0$ removes the marginal cost and recovers an RD entropy-majorization update. Perception enters through the signed multiplier and the reconstruction marginal retained for subsequent mixing.

For each positive-probability decoder state $s = (\mathbf{w}, y)$, let $\mu_s = \mathbb{P}\{s\}$, $\pi_s = \mathbb{P}\{T=1 \mid s\}$, and $\theta_s = q_d(1 \mid s)$. Then

$$D = a + \sum_s \mu_s(1-2\pi_s)\theta_s, \qquad b = \sum_s \mu_s \theta_s. \tag{12}$$

Set $\theta_s = 1$ if $\lambda(1-2\pi_s) + \nu < 0$, and zero if it is positive. At equality choose $1/2$. Zero-mass states can be assigned arbitrarily. At prescribed b, set $z_s = \mu_s\theta_s$ and solve

$$\begin{aligned} \min_{\boldsymbol{z}} \quad & a + \sum_s (1-2\pi_s) z_s \\ \text{s.t.} \quad & 0 \le z_s \le \mu_s, \qquad \sum_s z_s = b. \end{aligned} \tag{13}$$

Moving mass to a larger π_s decreases distortion. Sorting K positive-mass states in decreasing posterior order therefore gives an optimum with at most one fractional state, even with ties, in $O(K \log K)$ time. Cumulative allocations and their mixtures trace the entire fixed-encoder (D,b) frontier with unchanged rate. Algorithm 1 summarizes the resulting component search.

Proposition 1. *For fixed finite auxiliary alphabets and multipliers, sequential updates* (11) *followed by the decoder update have nonincreasing objective values, which converge in exact arithmetic.*

Proof. Adding the cost and fixed terms to (10) gives a tight upper bound U_i on L, and $L(q_i^+) \le U_i(q_i^+) \le U_i(q_i^{\mathrm{old}}) = L(q_i^{\mathrm{old}})$. Decoder minimization is exact and $L \ge -|\nu|$. Further derivation details and the posterior-allocation argument are in [19]. □

B. Marginal-Constrained Mixing and Refinement

For a pool $\mathcal{S}$ of J triples, solve

$$\begin{aligned} R_{\mathcal{S}}(D,P) = \min_{\boldsymbol{\eta}\ge 0} \quad & \sum_j \eta_j R_j \\ \text{s.t.} \quad & \sum_j \eta_j = 1, \quad \sum_j \eta_j D_j \le D, \\ & a - P \le \sum_j \eta_j b_j \le a + P. \end{aligned} \tag{14}$$

The weights specify predetermined subblock proportions. For fixed (R,P), define $D_{\mathcal{S}}(R,P)$ by minimizing $\sum_j \eta_j D_j$

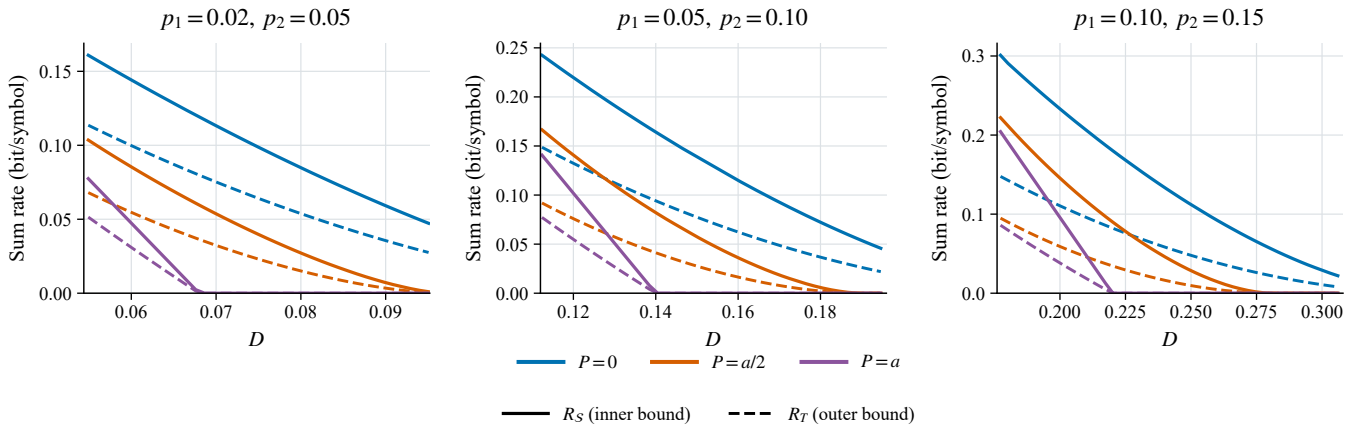


Fig. 2. Sum-rate boundaries of the RDP inner bound, $R_{\mathcal{S}}(D, P)$ (solid), and the cooperative outer bound, $R_T(D, P)$ (dashed), for the XOR cases in Table I. Colors identify $P \in \{0, a/2, a\}$. Curves connect evaluated values over $0.8a \le D \le 1.4a$.

Algorithm 1 BA-Type Component Search

Input: Source pmf, auxiliary alphabets, (λ, ν), positive local channels, tolerances, iteration cap

Initialize the decoder and evaluate L
for each sweep up to the cap **do**
  Store $(\boldsymbol{q}_{\text{old}}, L_{\text{old}}) \leftarrow (\boldsymbol{q}, L)$
  for $i = 1, \ldots, M$ **do**
    Compute ℓ_i, v_i and update q_i by (11)
  end for
  Minimize L over the decoder and reevaluate L
  If L increases numerically, restore the saved state and stop
  Stop if objective and channel-change tests both hold
end for
return Kernels and (R, D, b) from (7)

subject only to normalization, the marginal constraints, and $\sum_j \eta_j R_j \le R$. The distortion-budget constraint in (14) is removed. A rate budget need not be exhausted.

Using $\lambda, \alpha, \beta \ge 0$ for the budget constraints and $\tau \in \mathbb{R}$ for normalization, the LP dual is

$$\begin{aligned} \max_{\lambda,\alpha,\beta\ge 0,\ \tau\in\mathbb{R}} \quad & -\lambda D - (\alpha+\beta)P - \tau \\ \text{s.t.} \quad & R_j + \lambda D_j + (\alpha-\beta)(b_j - a) + \tau \ge 0, \\ & j \in [J]. \end{aligned} \tag{15}$$

The dual supplies $\nu = \alpha - \beta$ for channel search. A negative reduced cost violates a supporting inequality of the current pool. We explore multiple starts and auxiliary sizes, add posterior-frontier endpoints, and retain feasible nonworsening sequential quadratic programming (SQP) refinements [19]. A basic optimal mixture uses at most four components, which does not bound $|\mathcal{W}_i|$.

With R_{in} the infimum sum rate in Theorem 1, the retained strategies and the cooperative outer bound give

$$R_{T|Y}(D, P) \le R^\star(D, P) \le R_{\text{in}}(D, P) \le R_{\mathcal{S}}(D, P). \tag{16}$$

Here, $R^\star(D, P)$ is the operational RDP function, $R_{\text{in}}(D, P)$ is the sum-rate boundary of the single-letter inner bound, and $R_{\mathcal{S}}(D, P)$ is its finite-pool approximation. Thus, R_{in} and $R_{\mathcal{S}}$ describe the inner bound, whereas $R_{T|Y}$ describes the cooperative outer bound. No independence between T and Y is required. The gap cannot be uniquely decomposed from these computations. It contains the cooperative relaxation and finite-pool search error, and it may also include a gap between the operational RDP function and its product-auxiliary characterization. Objective convergence does not certify a global optimum of the nonconvex channel problem.

V. Numerical Results

For the XOR examples, let $Y \sim \text{Bern}(1/2)$ and independent $N_i \sim \text{Bern}(p_i)$, independent of Y. Set $X_i = Y \oplus N_i$ and $T = X_1 \oplus X_2$. Then $a = p_1 + p_2 - 2p_1p_2$ and $T \perp Y$, while observations are independent given Y. The lossless-description rate is $R_{\text{loss}} = h_2(p_1) + h_2(p_2)$, where h_2 is binary entropy. The exact zero-message distortion is

$$D_0(P) = a + (1 - 2a)\max\{0, a - P\}. \tag{17}$$

The saved pools include this endpoint and lossless descriptions. The search combines multiple starts over a multiplier grid, LP-dual directions, and auxiliary sizes two through four. It stops after two consecutive rounds with maximum rate gains below 0.0015 bit/symbol over the probe budgets, or after five rounds. Full parameters, seeds, and validation are recorded in [19].

Figure 2 brackets the operational rate at each tested pair. Across the three listed cases, G increases with a and is largest at $P = 0$ near $D = 0.8a$, reaching 0.1554 bit/symbol in the third case. This gap combines the cooperative relaxation and incomplete channel search, and it may also reflect a gap between the operational RDP function and its product-auxiliary characterization for the particular source model. The controlled tests below show only modest gains from

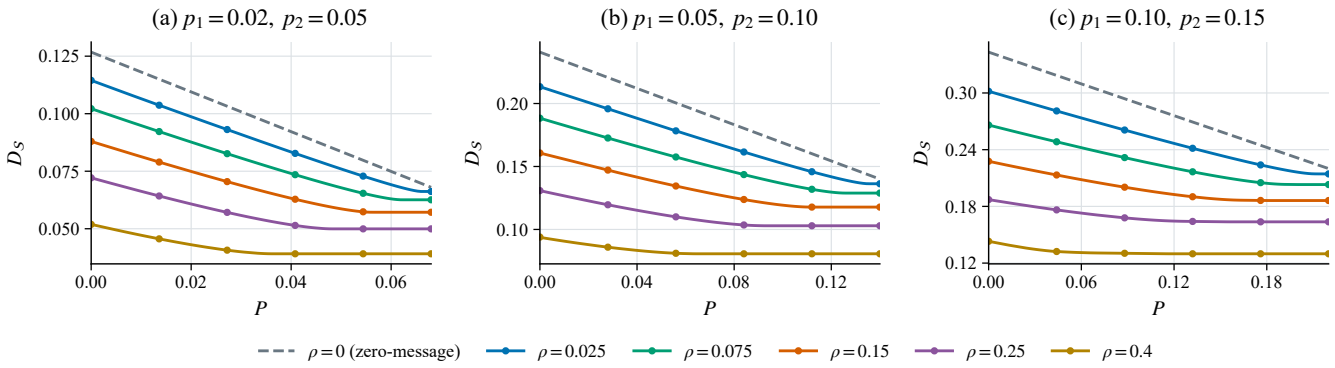


Fig. 3. Fixed-rate XOR sections $D_S(R, P)$ at $R = \rho R_{\text{loss}}$. The dashed $\rho = 0$ curve is (17). Markers identify selected saved LP grid values.

TABLE I
XOR POOLS AND SAMPLED RATE DIFFERENCES (BITS/SYMBOL): $\Delta_P = \max_D [R_S(D, 0) - R_S(D, a)]$ AND $G = \max_{D,P} [R_S - R_T]$ OVER SAVED BUDGETS WITH $D/a \in [0.8, 1.4]$ AND $P/a \in [0, 1]$.

p_1	p_2	a	J	Δ_P	G
0.02	0.05	0.068	493	0.1192	0.0475
0.05	0.10	0.140	623	0.1642	0.0910
0.10	0.15	0.220	678	0.1806	0.1554

additional search directions. The values Δ_P measure a finite-pool achievable rate reduction, not an operational converse penalty. For binary Hamming loss, $P = a$ recovers RD because empirical perception is at most distortion plus vanishing source-empirical fluctuation, and constant-zero reconstruction handles $D \geq a$ [19]. In each case, 41 direct RD LPs agree with the same-pool $P = a$ values within 1.6×10^{-8} bit/symbol.

Figure 3 shows the distortion benefit at fixed rate. Piecewise-linear sections arise from finite-pool mixing, and plateaus indicate inactive perception constraints. Three-dimensional XOR surfaces appear in [19]. Independent enumeration reproduces all archived metrics within 5.6×10^{-16}, with maximum budget excess 2.8×10^{-10}. At 27 single-terminal budgets, independent channel optimization and BA checks agree with the binary closed form to within 3.0×10^{-13} bit/symbol. These tests validate a known convex special case, rather than global distributed optimization.

Controlled experiments on coarse 40-direction pools show that six LP-dual directions reduce the rate by at most 0.01123 bit/symbol, whereas paired SQP refinement contributes at most 1.58×10^{-5} bit/symbol. Neither the dual nor matched fixed directions are uniformly better. Adding size-four auxiliary searches still lowers the rate by 0.00368 bit/symbol in the first case, which motivates searches beyond binary auxiliaries without isolating cardinality from additional search. Full protocols appear in [19].

For $T = X_1 \wedge X_2$ at $(p_1, p_2) = (0.10, 0.15)$, the target has $a = 0.39$ and is correlated with Y. Figure 4 compares the sum-rate boundary of the RDP inner bound with the conditional cooperative outer bound (5). Relaxing the perception budget from

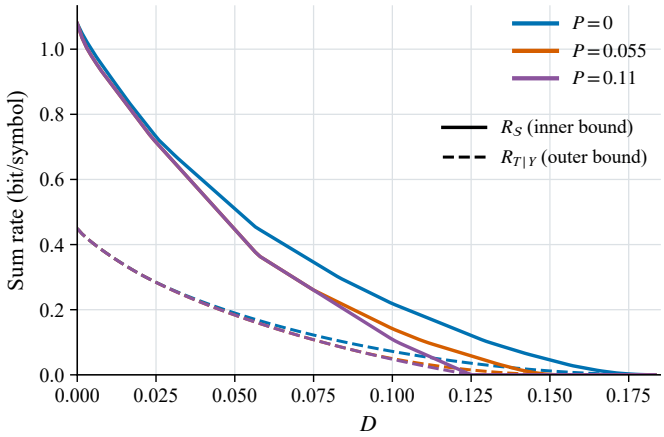


Fig. 4. Sum-rate boundaries of the RDP inner bound, $R_S(D, P)$ (solid), and the conditional cooperative outer bound, $R_{T|Y}(D, P)$ (dashed), for the AND target with $(p_1, p_2) = (0.10, 0.15)$ and $a = 0.39$.

$P = 0$ to $P = 0.11$ reduces the exact zero-message distortion from 0.1833 to 0.125. Unlike the XOR case, a reconstruction marginal above a can help, and both signs of the marginal multiplier are used in the search. At $(D, P) = (0.06, 0)$, the computed inner-bound rate is 0.4336 bit/symbol, compared with the outer-bound value 0.1603 bit/symbol. The fixed-rate sections and marginal-cancellation example are given in [19].

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VI. CONCLUSION

We computed the RDP function for distributed function coding with decoder side information using signed-marginal BA updates, posterior-sorted decoder frontiers, and marginal-aware LP mixing. The XOR results quantify the rate benefit of relaxing perception, while the AND result shows how target–side-information dependence changes the useful reconstruction marginal. The remaining inner–outer gap motivates stronger converses and structured codes. Extending the method beyond binary targets requires an LP over full reconstruction pmfs.

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APPENDIX A
CODING CONSTRUCTION AND AUXILIARY RESTRICTIONS

A. Empirical Perception and Achievability

We detail the empirical-distribution distinction and complete the proof of Theorem 1.

Jensen's inequality gives $d_{\mathrm{TV}}(p_T, \mathbb{E}[\hat{p}_{\hat{T}^n}]) \le P_n$, but fair coordinate marginals alone do not control P_n: repeating one fair bit produces an empirical pmf at distance $1/2$ from a fair binary target. Thus (1) constrains empirical statistics rather than the full block distribution.

In particular, P_n is not $d_{\mathrm{TV}}(p_{\hat{T}^n}, p_T^{\otimes n})$. Matching a one-letter marginal, controlling a sample's empirical distribution, and matching an entire block law are different requirements. The achievability argument below proves empirical convergence for the constructed codes, which is the extra step needed beyond checking the one-letter marginal. Decoder private randomness is allowed, but no shared random seed is required by that construction.

Proof. For fixed $Q = q$, the Berger–Tung covering, binning, and joint decoding construction [4], [5], with Y^n at the decoder, recovers jointly typical auxiliaries when the rate inequalities have slack. The decoder draws symbols according to $q_d(\cdot \mid \mathbf{w}, y, q)$. Conditional typicality gives convergence in probability of sample distortion and the empirical reconstruction pmf to their single-letter values. Boundedness yields convergence of expectations. Averaging over codebooks then gives a sequence of deterministic encoders with vanishing failure probability.

For finite Q, concatenate predetermined subblocks with $n_q/n \to p_Q(q)$. Time sharing among fixed-q inner-bound corner points yields (3), and the full-block empirical pmf converges to $\bar{p}_{\hat{T}} = \sum_q p_Q(q) p_{\hat{T}|Q=q} = p_{\hat{T}}$. The triangle inequality bounds the difference between expected empirical loss and $d_{\mathrm{TV}}(p_T, \bar{p}_{\hat{T}})$ by $\mathbb{E}[d_{\mathrm{TV}}(\hat{p}_{\hat{T}^n}, \bar{p}_{\hat{T}})] \to 0$. Expected distortion converges to $\mathbb{E}[d(T, \hat{T})]$. The known subblock schedule needs no common randomness, and closure removes the rate slack. □

Theorem 1 does not assume conditional independence. If $p_{\mathbf{X}|Y} = \prod_i p_{X_i|Y}$, the pairs (X_i, W_i) are independent given (Y, Q). Each subset mutual information in (3) then becomes $\sum_{i \in S} I(X_i; W_i \mid Y, Q)$, giving

$$R_i \ge I(X_i; W_i \mid Y, Q), \qquad i \in [M]. \tag{18}$$

B. Proof of the Inner–Outer Separation

Consider the RD inner and outer bounds in [14, Cor. 1 and Th. 2]. Let $\mathcal{R}_{\mathrm{a}}(D)$ denote the inner bound with product auxiliaries and $\mathcal{R}_{\mathrm{o}}(D)$ the stated outer bound with individual Markov conditions. Both require a decoder $g(\mathbf{W}, Y)$ with $\mathbb{E}[d(T, g(\mathbf{W}, Y))] \le D$ and impose

$$\sum_{i \in S} R_i \ge I(\mathbf{X}; W_S \mid W_{S^c}, Y), \quad \emptyset \ne S \subseteq [M]. \tag{19}$$

The inner class requires $p_{\mathbf{W}|\mathbf{X}Y} = \prod_i p_{W_i|X_i}$, whereas the outer class requires only $W_i - X_i - (X_{-i}, Y)$ for each i. These individual conditions need not make the joint conditional distribution of $\mathbf{W}$ factor, even when the observations are independent given Y.

Proposition 2. *Conditional independence given Y does not ensure $\mathcal{R}_{\mathrm{a}}(D) = \mathcal{R}_{\mathrm{o}}(D)$. A three-encoder binary example has*

$$\inf_{\boldsymbol{R} \in \mathcal{R}_{\mathrm{o}}(0)} \sum_{i=1}^{3} R_i \le 2 < 3 = \inf_{\boldsymbol{R} \in \mathcal{R}_{\mathrm{a}}(0)} \sum_{i=1}^{3} R_i = R_{\mathrm{RD}}^{\star}(0). \tag{20}$$

The same example strictly separates the corresponding RDP inner and outer bounds at $(D, P) = (0, 0)$.

Proof. Let $M = 3$, let Y be constant, and take independent fair bits X_1, X_2, X_3. Set $T = X_1 \oplus X_2 \oplus X_3$, with $\oplus$ denoting modulo-two addition, and use Hamming loss $d(t, \hat{t}) = \mathbf{1}\{t \ne \hat{t}\}$. For an independent fair bit Z, choose

$$W_1 = 0, \qquad W_2 = Z, \qquad W_3 = Z \oplus T. \tag{21}$$

Each W_i is independent of $\mathbf{X}$ and meets its Markov condition, yet $W_2 \oplus W_3 = T$. The right-hand side of (19) is zero for $S = \{1\}$ and one for every other nonempty S. Hence $(0, 1, 1) \in \mathcal{R}_{\mathrm{o}}(0)$.

For any achievable code, reveal X_{-i}^n to the decoder. These blocks determine all messages except M_i, while $\hat{X}_{i,k} = \hat{T}_k \oplus \bigoplus_{j \ne i} X_{j,k}$ estimates $X_{i,k}$ with average error D_n. Coordinatewise Fano and concavity of binary entropy give

$$nR_i \ge I(X_i^n; M_i \mid X_{-i}^n) \ge n[1 - h_2(D_n)]. \tag{22}$$

Here $h_2(u) = -u \log_2 u - (1-u) \log_2 (1-u)$ is binary entropy, with $0 \log_2 0 = 0$. Independent decoder randomness does not change the conditional entropy in (22). As $D_n \to 0$, each rate is at least one bit, while product channels $W_i = X_i$ attain $(1, 1, 1)$. This proves (20) under closure and time sharing. Moreover, exact reconstruction gives $\hat{T}^n = T^n$. The law of large numbers then implies that the empirical distribution of $\hat{T}^n$ converges to p_T, and hence $P_n \to 0$. The same construction therefore strictly separates the corresponding RDP inner and outer bounds at $(D, P) = (0, 0)$. □

Here $R_{\mathrm{RD}}^{\star}$ denotes the operational sum rate without a perception constraint. The product-auxiliary inner bound is tight in this example, whereas the stated outer bound admits a smaller sum rate. The product replacement in [14, Th. 3, eqs. (26)–(27)] preserves each $p_{X_i W_i Y}$ but not the joint dependence among the auxiliaries. In the construction above, this dependence carries the parity information required for reconstruction. After the replacement, the minimum distortion increases to $1/2$. Thus preserving the individual marginals is insufficient to preserve the distortion constraint.

For computation, this example gives a concrete reason to distinguish the two auxiliary classes: a search over the larger class could return descriptions that separate local encoders cannot implement. We therefore compute achievable strategies using the product law in Theorem 1.

APPENDIX B
COOPERATIVE OUTER BOUND AND BINARY BENCHMARK

A. Conditional Cooperative Converse

The inner bound restricts each encoder to its own observation. Giving all observations and the side information to one encoder removes this restriction, and the cooperative encoder can compute $T = f(\mathbf{X}, Y)$. Define

$$\begin{aligned} R_{T|Y}(D,P) = \min_{q(\hat{t}|t,y)} \quad & I(T;\hat{T} \mid Y) \\ \text{s.t.} \quad & \mathbb{E}[d(T,\hat{T})] \le D, \\ & d_{\mathrm{TV}}(p_T, p_{\hat{T}}) \le P. \end{aligned} \tag{23}$$

The perception constraint is imposed on the unconditional reconstruction marginal, consistently with (1).

Proposition 3. *Every achievable distributed code under* (1) *satisfies* $R^\star(D,P) \ge R_{T|Y}(D,P)$*, without requiring* $T \perp Y$.

Proof. Write $\boldsymbol{M} = (M_1, \dots, M_M)$ for all messages. For a blocklength-n code with rates at most $R_i + \epsilon$, message counting and data processing give

$$\begin{aligned} n\sum_i (R_i + \epsilon) &\ge H(\boldsymbol{M}) \ge I(\mathbf{X}^n; \boldsymbol{M} \mid Y^n) \\ &\ge I(T^n; \boldsymbol{M} \mid Y^n) \ge I(T^n; \hat{T}^n \mid Y^n). \end{aligned} \tag{24}$$

The third inequality uses the fact that T^n is a function of $(\mathbf{X}^n, Y^n)$. Independent decoder randomness preserves the Markov chain $T^n - (\boldsymbol{M}, Y^n) - \hat{T}^n$, which gives the last inequality.

Let K be uniform on $[n]$ and independent of all other variables. Since the pairs (T_k, Y_k) are i.i.d., conditioning and the chain rule imply

$$\begin{aligned} I(T^n; \hat{T}^n \mid Y^n) &\ge \sum_{k=1}^n I(T_k; \hat{T}_k \mid Y_k) \\ &= nI(T_K; \hat{T}_K \mid Y_K, K) \\ &\ge nI(T_K; \hat{T}_K \mid Y_K). \end{aligned} \tag{25}$$

For the first inequality, expand the first entropy term using the i.i.d. pairs (T_k, Y_k) and upper-bound the second by subadditivity and removal of conditioning. The last inequality follows because (T_K, Y_K) is independent of K. The induced channel from (T_K, Y_K) to $\hat{T}_K$ has distortion D_n, and Jensen's inequality gives

$$d_{\mathrm{TV}}(p_T, p_{\hat{T}_K}) = d_{\mathrm{TV}}(p_T, \mathbb{E}[\hat{p}_{\hat{T}^n}]) \le P_n. \tag{26}$$

Thus this channel is feasible for (23) at (D_n, P_n). For each fixed $\epsilon > 0$, take a convergent subsequence of the averaged channels, which exists because the alphabets are finite. Distortion, total variation, and mutual information are continuous on this compact set. The limiting channel meets (D, P) and has conditional mutual information at most $\sum_i R_i + M\epsilon$. Hence $R_{T|Y}(D,P) \le \sum_i R_i + M\epsilon$. Letting $\epsilon \downarrow 0$ and taking the infimum over achievable rate vectors proves the claim. □

When $T \perp Y$, $I(T;\hat{T} \mid Y) = I(T;\hat{T}, Y) \ge I(T;\hat{T})$. Conversely, a channel that ignores Y attains the unconditional information RDP function. Hence $R_{T|Y}(D,P) = R_T(D,P)$ in the XOR examples, where

$$\begin{aligned} R_T(D,P) = \min_{q(\hat{t}|t)} \quad & I(T;\hat{T}) \\ \text{s.t.} \quad & \mathbb{E}[d(T,\hat{T})] \le D, \\ & d_{\mathrm{TV}}(p_T, p_{\hat{T}}) \le P. \end{aligned} \tag{27}$$

B. Closed-Form Evaluation for a Binary Target

For $T \sim \mathrm{Bern}(a)$, $0 < a < 1/2$, and Hamming distortion, set $\bar{P} = \min\{P, a\}$ and define

$$D_0 = a + (1-2a)(a - \bar{P}), \qquad D_1 = \frac{\bar{P}}{1 - 2(a - \bar{P})}. \tag{28}$$

The binary RDP expression of [1] can be written as

$$R_T(D,P) = \begin{cases} 0, & D \ge D_0, \\ h_2(a) - h_2(D), & 0 \le D \le D_1, \\ h_2(a) + h_2(a-\bar{P}) & \\ \qquad - H_4(\boldsymbol{j}), & D_1 < D < D_0, \end{cases} \tag{29}$$

where $H_4(\boldsymbol{j}) = -\sum_{u,v\in\{0,1\}} j_{uv} \log_2 j_{uv}$ is the entropy of the joint pmf

$$\begin{aligned} j_{01} &= (D - \bar{P})/2, \quad j_{10} = (D + \bar{P})/2, \\ j_{00} &= 1 - a - j_{01}, \quad j_{11} = a - j_{10}. \end{aligned} \tag{30}$$

The first subscript is T and the second is $\hat{T}$. Only the active-perception branch uses (30). Its reconstruction marginal is $a - \bar{P}$ and its distortion is D. The parametrization below explains the thresholds and supplies an independent validation.

Together with the achievable construction, Proposition 3 gives

$$R_{T|Y}(D,P) \le R^\star(D,P) \le R_{\mathrm{in}}(D,P) \le R_{\mathcal{S}}(D,P). \tag{31}$$

Here R_{in} is the infimum sum rate in $\mathcal{R}_{\mathrm{in}}(D,P)$, and $R_{\mathcal{S}}$ is its finite-strategy approximation defined below. The separation between the two computable endpoints cannot be uniquely decomposed. It contains the cooperative relaxation and finite-pool search error, and it may also include a gap between the operational RDP function and its product-auxiliary characterization.

C. Independent Single-Terminal Validation

Write $u = \mathbb{P}\{\hat{T} = 1 \mid T = 0\}$ and $v = \mathbb{P}\{\hat{T} = 0 \mid T = 1\}$. For a binary channel,

$$\begin{aligned} b &= (1-a)u + a(1-v), \\ D &= (1-a)u + av, \\ I(T;\hat{T}) &= h_2(b) - (1-a)h_2(u) - ah_2(v). \end{aligned} \tag{32}$$

This gives an independent two-variable convex program for (27), with affine distortion and marginal constraints and bounds $0 \le u, v \le 1$. Our XOR outer bounds use the closed form, not the output of this minimizer.

Without perception, the positive-rate binary RD channel has reverse form $T = \hat{T} \oplus E$, where $E \sim \mathrm{Bern}(D)$ is independent of $\hat{T}$. It has rate $h_2(a) - h_2(D)$ and marginal $b = (a - D)/(1 - 2D)$ for $D < a$. Its mismatch reaches P when $D = D_1$ in (28). Beyond this threshold, the optimal feasible marginal is $b = a - \bar{P}$, which gives (30). At zero mutual information the reconstruction is independent of the target, and the smallest distortion under perception is D_0. These are the three branches of (29).

The supporting multipliers provide a second check. For a strictly positive joint pmf with fixed D and b, put

$$\lambda = -\tfrac{1}{2}\log_2 \frac{j_{01}j_{10}}{j_{00}j_{11}},$$
$$\nu = -\log_2 \frac{1-b}{b} - \tfrac{1}{2}\log_2 \frac{j_{01}j_{11}}{j_{00}j_{10}}. \tag{33}$$

These are the negatives of the derivatives of I with respect to D and b. On the active-perception branch, $\lambda \geq 0$ and $\nu \leq 0$, as required by the lower-marginal constraint. On the RD branch, $\nu = 0$. Convexity and these supporting conditions verify the minimizer in the positive-rate interior, and continuity supplies the boundary values.

Let $c(t, \hat{t}) = \lambda \mathbf{1}\{t \neq \hat{t}\} + \nu \hat{t}$. Starting with a positive $q(\hat{t} \mid t)$, the one-terminal BA iteration is

$$q^+(\hat{t} \mid t) = \frac{p_{\hat{T}}^{\mathrm{old}}(\hat{t}) 2^{-c(t,\hat{t})}}{\sum_{z \in \{0,1\}} p_{\hat{T}}^{\mathrm{old}}(z) 2^{-c(t,z)}}. \tag{34}$$

The 27 tests use (33) and stop when the largest channel-entry change is below 10^{-13}, with a cap of 200000 iterations. They verify a supporting objective as well as rate, distortion, and marginal, rather than infer correctness from a visually smooth curve.

D. Conditional Binary Evaluation

For binary T and Y, write $\omega_y = \mathbb{P}\{Y = y\}$, $a_y = \mathbb{P}\{T = 1 \mid Y = y\}$, and

$$u_y = \mathbb{P}\{\hat{T} = 1 \mid T = 0, Y = y\},$$
$$v_y = \mathbb{P}\{\hat{T} = 0 \mid T = 1, Y = y\}. \tag{35}$$

With $b_y = (1 - a_y)u_y + a_y(1 - v_y)$, the conditional outer bound is the four-variable convex program

$$\begin{aligned} \min_{0 \leq u_y, v_y \leq 1} \quad & \sum_y \omega_y \{h_2(b_y) - (1 - a_y)h_2(u_y) - a_y h_2(v_y)\} \\ \text{s.t.} \quad & \sum_y \omega_y [(1 - a_y)u_y + a_y v_y] \leq D, \\ & a - P \leq \sum_y \omega_y b_y \leq a + P, \end{aligned} \tag{36}$$

where $a = \sum_y \omega_y a_y$. Mutual information is convex in the conditional channel for fixed p_{TY}, and all constraints are affine.

For the AND example, $(a_0, a_1) = (0.015, 0.765)$ and $\omega_0 = \omega_1 = 1/2$. We solve (36) from deterministic feasible starts formed by mixing perfect recovery with zero-rate channels.

Algorithm 2 BA-Type Algorithm for a Component Strategy

Input: $p_{T\mathbf{X}Y}$, $(\mathcal{W}_i)$, (λ, ν), positive (q_i), tolerances $(\varepsilon_L, \varepsilon_q, \varepsilon_{\mathrm{num}})$, iteration cap N_{iter}
Output: Local channels, decoder, and achieved (R, D, b)

```
Initialize q_d by (43) and evaluate L by (37)
for u = 1, ..., N_iter do
    q_old ← q and L_old ← L
    for i = 1, ..., M do
        Set q_i^old ← q_i
        compute v_i, ℓ_i by (39) and (38), respectively
        update q_i by (41)
    end for
    Update q_d by (43) and reevaluate L
    if L − L_old > ε_num(1 + |L_old|) then
        Restore (q, L) ← (q_old, L_old) and break
    end if
    if both objective and channel-change tests hold then
        break
    end if
end for
return q and its metrics from (7)
```

Over 363 plotted budgets, the largest constraint residual is 1.1×10^{-12}, both bounds are nonincreasing in D to numerical precision, and the conditional outer bound never exceeds the saved inner bound.

Appendix C
Computational Derivations and Implementation

The component metrics are (7). Algorithm 2 expands Algorithm 1 with explicit stopping and numerical safeguards. Its local-channel and decoder updates are derived below.

A. Optimization of the Local Test Channels

With mixture weights $\eta_j \geq 0$, $\sum_j \eta_j = 1$, rates, distortions, and marginals average linearly. Perception loss is $|\sum_j \eta_j b_j - a|$: components with marginals $a \pm \epsilon$ cancel when mixed equally. We therefore search using

$$L_{\lambda,\nu}(\boldsymbol{q}) = R + \lambda D + \nu(b - a), \tag{37}$$

where $\lambda \geq 0$, $\nu \in \mathbb{R}$, and L denotes this objective. Multipliers $\alpha, \beta \geq 0$ on $b - a \leq P$ and $a - b \leq P$ give $\nu = \alpha - \beta$, with budget-only terms omitted.

To update one encoder, fix the decoder and every local test channel except q_i, and define the effective cost

$$\ell_i(x_i, w_i) = \mathbb{E}[\lambda d(T, \hat{T}) + \nu \hat{T} \mid X_i = x_i, W_i = w_i], \tag{38}$$

The expectation uses the source, other channels, and fixed decoder. Conditioning on (X_i, W_i) removes the trial factor $q_i(w_i \mid x_i)$ in (2), leaving ℓ_i fixed during the update. For the current positive channel q_i^{old}, set

$$v_i(w_i \mid y) = \sum_{x_i} p_{X_i|Y}(x_i \mid y) q_i^{\mathrm{old}}(w_i \mid x_i). \tag{39}$$

For trial q_i, let $p^{q_i}_{W_i|Y}(w \mid y) = \sum_x p_{X_i|Y}(x \mid y) q_i(w \mid x)$ denote its induced marginal. Holding v_i fixed gives

$$\begin{aligned}\sum_{x_i,y,w_i} & p_{X_iY}(x_i,y) q_i(w_i \mid x_i) \log_2 \frac{q_i(w_i \mid x_i)}{v_i(w_i \mid y)} \\ &= I_{q_i}(X_i; W_i \mid Y) \\ &\quad + \mathbb{E}_Y\left[\mathrm{D_{KL}}\left(p^{q_i}_{W_i|Y}(\cdot \mid Y) \| v_i(\cdot \mid Y)\right)\right], \end{aligned} \tag{40}$$

Here I_{q_i} uses the trial channel and $\mathrm{D_{KL}}(p\|q) = \sum_w p(w) \log_2[p(w)/q(w)]$ is Kullback–Leibler divergence. Its nonnegative term vanishes at q_i^{old}, where the induced marginal equals v_i. Adding (38) to this tight upper bound and minimizing rowwise gives

$$\begin{aligned}\psi_i(x_i, w_i) &= \sum_y p_{Y|X_i}(y \mid x_i) \log_2 v_i(w_i \mid y) - \ell_i(x_i, w_i), \\ q_i^+(w_i \mid x_i) &= \frac{2^{\psi_i(x_i,w_i)}}{\sum_{w_i' \in \mathcal{W}_i} 2^{\psi_i(x_i,w_i')}}. \end{aligned} \tag{41}$$

ψ_i is the base-two log-weight. Conditional independence makes other encoders' rates independent of q_i. A sweep updates local channels sequentially and then the decoder. We use log-domain weights and omit zero-probability source letters.

Setting $\nu = 0$ removes the output-marginal cost and recovers an RD entropy-majorization update with decoder side information. The objective then has the conditional-mutual-information form used in the distributed RD method of Tang and Yang [14]. Our derivation makes the conditional expected cost explicit through (38). The RDP-specific additions are the signed marginal multiplier, the retention of b during mixture optimization, and the supporting directions obtained from the perception-constrained LP. The BA construction itself is not a new principle.

At each sweep's start, $\boldsymbol{q}_{\mathrm{old}}$ and L_{old} store the pre-update channels and objective value. The output (R, D, b) then enters the mixture optimization.

B. Decoder Optimization and Descent

For positive-probability decoder states $s = (\mathbf{w}, y)$, let $\mu_s = \mathbb{P}\{(\mathbf{W}, Y) = s\}$, $\pi_s = \mathbb{P}\{T = 1 \mid s\}$, and $\theta_s = q_d(1 \mid s)$. Then

$$D = a + \sum_s \mu_s (1 - 2\pi_s)\theta_s, \qquad b = \sum_s \mu_s \theta_s. \tag{42}$$

Linearity in θ_s makes the fixed-channel decoder optimum

$$\theta_s = \begin{cases} 1, & \lambda(1 - 2\pi_s) + \nu < 0, \\ 0, & \lambda(1 - 2\pi_s) + \nu > 0, \end{cases} \tag{43}$$

At equality, any $\theta_s \in [0, 1]$ is optimal, and we choose $1/2$. Also $q_d(0 \mid s) = 1 - \theta_s$, and zero-mass states can be assigned arbitrarily.

Proposition 4. *For fixed finite auxiliary alphabets and (λ, ν), successive updates* (41) *and* (43) *do not increase L in exact arithmetic, and their objective values converge.*

Proof. Adding the linear cost and terms independent of q_i to (40) gives an upper bound U_i on L that is tight at q_i^{old}. With the other kernels fixed, $L(q_i^+) \le U_i(q_i^+) \le U_i(q_i^{\mathrm{old}}) = L(q_i^{\mathrm{old}})$. The subsequent decoder update also minimizes L for the fixed local channels, and the lower bound $L \ge -|\nu|$ therefore ensures convergence of the objective values. □

At prescribed marginal b, set $z_s = \mu_s \theta_s$. Equation (42) gives the fixed-channel linear problem

$$\begin{aligned} \min_{\boldsymbol{z}} \quad & a + \sum_s (1 - 2\pi_s) z_s \\ \text{s.t.} \quad & 0 \le z_s \le \mu_s, \qquad \sum_s z_s = b, \end{aligned} \tag{44}$$

where $\boldsymbol{z} = (z_s)_s$. Moving allocation from a smaller to a larger π_s decreases distortion. Thus sorting by decreasing posterior gives an optimum with at most one fractional state, even with ties. Sorting K states costs $O(K \log K)$, and cumulative allocations and their mixtures trace the fixed-encoder (D, b) frontier at unchanged rate.

The fractional-state statement concerns a fixed marginal b. In contrast, the tie choice $\theta_s = 1/2$ in (43) concerns a fixed multiplier pair. Multiple tied states may use that value without affecting the Lagrangian optimum.

After each sweep, Algorithm 2 stops early only if both $|L - L_{\mathrm{old}}| \le \varepsilon_L(1 + |L|)$ and $\|\boldsymbol{q} - \boldsymbol{q}_{\mathrm{old}}\|_\infty \le \varepsilon_q$. Proposition 4 concerns exact-arithmetic objective convergence, whereas these tests govern finite-iteration stopping.

C. Mixture Optimization and Refinement

Let $\mathcal{S}$ contain J component triples (R_j, D_j, b_j). For budgets (D, P), optimize mixture weights $\boldsymbol{\eta} = (\eta_1, \ldots, \eta_J)$ by

$$\begin{aligned} R_{\mathcal{S}}(D, P) = \min_{\boldsymbol{\eta} \ge 0} \quad & \sum_j \eta_j R_j \\ \text{s.t.} \quad & \sum_j \eta_j = 1, \quad \sum_j \eta_j D_j \le D, \\ & a - P \le \sum_j \eta_j b_j \le a + P. \end{aligned} \tag{45}$$

The optimized η_j specify predetermined subblock proportions as in Theorem 1. At rate budget R, a second LP minimizes $\sum_j \eta_j D_j$ with the same normalization and marginal constraints and $\sum_j \eta_j R_j \le R$, defining $D_{\mathcal{S}}(R, P)$. The rate budget need not be exhausted.

For the finite LP, let $\lambda \ge 0$ multiply distortion, $\alpha, \beta \ge 0$ the two marginal inequalities, and $\tau \in \mathbb{R}$ normalization. A candidate has reduced cost $\bar{c}_j = R_j + \lambda D_j + \nu(b_j - a) + \tau$, where $\nu = \alpha - \beta$. Negative reduced cost violates a dual inequality and supplies multipliers for (37). We run Algorithm 2 from multiple starts and attempt joint refinement by sequential quadratic programming with analytic derivatives, accepting only feasible nonworsening strategies. We add decoder-frontier endpoints from (44), explore dual and extra directions and auxiliary sizes, and remove metric triples redundant in the convex hull.

For test budgets $\mathcal{B}$ and retained pool $\mathcal{S}_h$, measure $g_h = \max_{(D,P)\in\mathcal{B}}[R_{\mathcal{S}_h}(D,P) - R_{\mathcal{S}_{h+1}}(D,P)]$. The outer search stops after two small-gain rounds, separately from the channel-change tests. This finite-grid stability does not bound the gap to all product channels.

Let $R_{\mathrm{in}}(D,P) = \inf_{\boldsymbol{R}\in\mathcal{R}_{\mathrm{in}}(D,P)} \sum_i R_i$. Theorem 1 makes retained strategies and their predetermined mixtures achievable, giving

$$R^\star(D,P) \le R_{\mathrm{in}}(D,P) \le R_{\mathcal{S}}(D,P). \tag{46}$$

Enlarging $\mathcal{S}$ or relaxing a budget cannot increase $R_{\mathcal{S}}$. Stored kernels and weights specify the reported achievable strategies.

D. Interpreting the LP Dual

The full dual is (15) in the main text. Finite LP strong duality applies to the retained pool, and the coefficient $\nu = \alpha - \beta$ is the signed multiplier used in the BA objective. A negative reduced-cost component violates a dual inequality and supplies a missing supporting strategy. Failure of a local search to find one does not certify that none exists outside the pool. The four-component support bound follows from one normalization equality and at most three active budget inequalities. It does not bound the alphabet of any local auxiliary.

Appendix D
Experimental Protocols and Supplementary Results

We first revisit three binary XOR cases using the saved strategy pools. We then report independent single-terminal checks, controlled search experiments, and a newly computed AND example in which the target is correlated with the decoder side information. These experiments have separate data records and search protocols.

A. Binary XOR Model and the Archived Search

Set $M = 2$, $Y \sim \mathrm{Bern}(1/2)$, and independent $N_i \sim \mathrm{Bern}(p_i)$, independent also of Y. With $X_i = Y \oplus N_i$, the target $T = X_1 \oplus X_2 = N_1 \oplus N_2$ has $a = p_1 + p_2 - 2p_1p_2$. Thus X_1, X_2 are independent given Y, and $T \perp Y$. We use Hamming distortion and empirical total variation. Although Y provides no direct information about this target, it helps interpret each observation by removing their common binary mask. This structure makes a closed-form cooperative outer bound available.

Separate lossless descriptions yield $(R, D, P) = (R_{\mathrm{loss}}, 0, 0)$, where $R_{\mathrm{loss}} = h_2(p_1) + h_2(p_2)$. With no messages, $T \perp Y$ makes the reconstruction independent of T. A reconstruction marginal b therefore gives distortion $a + (1-2a)b$. Since $a < 1/2$, optimization over $|b - a| \le P$ gives

$$D_0(P) = a + (1-2a)\max\{0, a - P\}. \tag{47}$$

Independent reconstruction symbols with marginal $b = \max\{0, a - P\}$ attain this value and their empirical pmf concentrates. Both this endpoint and the lossless strategy are included in the saved pools. Boundary LPs are evaluated directly to avoid accepting a tiny positive information rate under a nominal zero-rate budget.

The archived search uses six starts and 18 logarithmically spaced distortion multipliers $\lambda \in [0.4, 40]$. Its seven initial ratios ν/λ are $-1, -0.8, -0.5, -0.2, 0, 0.5, 1$. Algorithm 2 uses at most 160 sweeps with $\varepsilon_L = \varepsilon_{\mathrm{num}} = 10^{-10}$ and $\varepsilon_q = 10^{-6}$. Sequential quadratic programming (SQP) uses at most 250 iterations with tolerance 10^{-11}. Outer refinement probes 54 budget pairs and auxiliary sizes two through four using LP dual and additional directions. It stops after two rounds with $g_h < 0.0015$ bit per symbol or five rounds. All three cases meet the former rule. Seed 20260914 fixes randomized starts.

Each case contains 4346 saved RDP grid values and 606 fixed-rate DP values, with kernels and mixture weights. The main comparisons below reuse those strategies unchanged. New outer-bound columns and evaluation grids are computed separately, without fitting the original values.

B. Perception Relaxation and the RD Limit

For binary Hamming distortion, $P = a$ suffices to recover the ordinary RD optimum, irrespective of whether T and Y are independent.

Proposition 5. *Let $T \sim \mathrm{Bern}(a)$ with $0 < a \le 1/2$. Under (1), $R^\star(D,a) = R^\star_{\mathrm{RD}}(D)$. The analogous equality holds between the finite-pool RDP and RD LPs whenever the pool contains the constant-zero reconstruction strategy.*

Proof. For any coupled binary $T, \hat{T}$, $|\mathbb{P}\{\hat{T} = 1\} - a| \le \mathbb{P}\{T \ne \hat{T}\}$. At blocklength n, applying the triangle inequality to empirical pmfs gives

$$P_n \le D_n + \mathbb{E}[d_{\mathrm{TV}}(\hat{p}_{T^n}, p_T)]. \tag{48}$$

The second term tends to zero for the i.i.d. source. If $D \le a$, every RD-achieving sequence therefore meets perception budget a. If $D \ge a$, the constant-zero decoder has zero rate, distortion a, and perception loss a. These two cases prove the operational equality. In a finite pool, the one-letter inequality applies to every mixture. The same split at $D = a$, using the constant-zero strategy, proves the LP equality. □

This argument avoids assuming that all RD-optimal decoder marginals lie below a. For the three saved XOR pools, 41 direct RD LPs per case agree with their $P = a$ values to within 1.6×10^{-8} bit per symbol. This is a same-pool RD consistency check, not a comparison against an assumed exact distributed RD formula.

C. Inner–Outer Comparisons and Fixed-Rate Sections

Table II lists source parameters, the lossless-description rate, pool size, and two different rate differences. The quantity Δ_P compares inner-bound rates at $P = 0$ and $P = a$ using the same pool. The quantity G compares the inner bound with the cooperative outer bound. Both maxima use the archived budget grid restricted to $0.8a \le D \le 1.4a$ and $0 \le P \le a$.

TABLE II

XOR CASES AND COMPUTED RATE DIFFERENCES (BITS/SYMBOL). HERE $\Delta_P = \max_{D \in \mathcal{D}_{\mathrm{grid}}} [R_{\mathcal{S}}(D,0) - R_{\mathcal{S}}(D,a)]$ AND $G = \max_{(D,P) \in \mathcal{B}_{\mathrm{grid}}} [R_{\mathcal{S}}(D,P) - R_T(D,P)]$. THE SETS USE ARCHIVED GRID BUDGETS IN $0.8a \le D \le 1.4a$ AND $0 \le P \le a$.

p_1	p_2	a	R_{loss}	J	Δ_P	G	D/a at max gap	P/a at max gap
0.02	0.05	0.068	0.428	493	0.1192	0.0475	0.803	0
0.05	0.10	0.140	0.755	623	0.1642	0.0910	0.826	0
0.10	0.15	0.220	1.079	678	0.1806	0.1554	0.801	0

TABLE III

NEW CONTROLLED EXPERIMENTS: MAXIMUM RATE REDUCTION OVER 54 PROBE BUDGETS (BITS/SYMBOL). DUAL AND FIXED EXTENSIONS EACH ADD SIX DIRECTIONS. THE SQP COLUMN COMPARES BA PLUS DUAL DIRECTIONS AGAINST THE SAME POOL AUGMENTED BY THEIR PAIRED SQP OUTPUTS. ALPHABET COLUMNS USE CUMULATIVE POOLS ON THE SAME 40-DIRECTION SWEEP.

p_1	p_2	Dual extension	Fixed extension	SQP after dual	$K \le 2$ to $K \le 3$	$K \le 3$ to $K \le 4$
0.02	0.05	0.00252	0.01118	6.00×10^{-7}	6.52×10^{-9}	0.00368
0.05	0.10	0.00624	0.00469	1.54×10^{-6}	0.00123	1.21×10^{-5}
0.10	0.15	0.01123	0.01088	1.58×10^{-5}	5.01×10^{-8}	1.15×10^{-8}

Figure 2 displays three perception budgets. At a fixed budget pair, (31) brackets the operational sum rate between the solid inner-bound and dashed outer-bound curves. The largest sampled gaps occur at perfect perception and near the left edge of the reported distortion window. Across these three cases, the larger-a examples leave a wider gap between the inner and cooperative outer bounds. Its maximum over this finite grid also bounds $R_{\mathcal{S}} - R^\star$ from above at the tested pairs, rather than providing a uniform guarantee between them. The saved three-dimensional views are retained in Appendix D.

The values of Δ_P show how the searched strategies benefit from perception relaxation. They are finite-pool rate differences, not converse bounds on the operational perception penalty. Figure 3 offers the complementary view at fixed rate: relaxing P lowers the minimum achievable distortion, especially at low positive rates. Here R is a budget and need not be exhausted. Lines join saved LP values, and plateaus indicate that perception is no longer active for the best mixture in the pool. Piecewise-linear sections are expected from finite-strategy mixing.

D. Single-Terminal and Numerical Checks

We independently test the binary formula (29) at 27 positive-rate budget pairs. These use $a \in \{0.068, 0.14, 0.22\}$, $P/a \in \{0, 1/2, 1\}$, and $D/D_0(P) \in \{0.2, 0.55, 0.85\}$. A two-variable constrained channel optimization agrees with the closed-form rates to within 2.1×10^{-13} bit per symbol. A single-terminal BA iteration with the corresponding signed-marginal supporting multipliers agrees in rate to within 3.0×10^{-13} and in Lagrangian value to within 3.4×10^{-16}. The channel parametrization and multipliers are given in Appendix B. These checks test a known convex special case independently of the distributed search.

All archived kernels and mixture weights are replayed by scalar probability enumeration. Recomputed metrics differ by at most 5.6×10^{-16}, with maximum original budget excess 2.8×10^{-10}. Monotonicity, zero-message, and inverse-budget checks also pass. Detailed numerical residuals and CSV hashes accompany the source package.

E. Controlled Alphabet and Search Refinement Experiments

Intermediate pre-SQP kernels were not retained in the archived main search. We therefore run new controlled experiments rather than infer an ablation from its aggregate gains. Each case uses 40 initial multiplier pairs: eight logarithmically spaced $\lambda \in [0.4, 40]$ and five ratios $\nu/\lambda \in \{-1, -0.5, 0, 0.5, 1\}$. There are four starts, the same iteration tolerances as above, and 54 probe budgets with nine distortion budgets from $0.25a$ to $D_0(0)$ and six perception budgets from zero to a. Seeds are 20261009, 20361009, and 20461009 for the three cases.

For the binary-auxiliary BA/SQP comparison, we record paired pre- and post-SQP strategies. BA-only pools retain the pre-SQP endpoints, and SQP-augmented pools retain both sets. Because a lower weighted objective does not imply improvement at every budget, replacing the former point would confound the comparison. All pools include the analytic baselines and are completed by posterior-sorted decoder frontiers. We compare the initial BA pool, that pool plus six LP-dual directions, and the latter augmented by the paired SQP outputs. A control adds six predetermined directions with λ logarithmically spaced in $[0.6, 30]$ and $\nu = -0.45\lambda$. It uses the same start count and seed labels as the dual extension.

Table III shows that dual directions improve the coarse pool, with maximum gains at $(D/a, P/a) = (0.25, 0)$, $(0.434, 0.2)$, and $(0.905, 0)$, respectively. They are not uniformly superior to the matched fixed-direction extension. SQP provides much smaller additional gains in these runs. Most of the observed refinement benefit comes from exploring further supporting directions rather than from joint polishing.

For alphabet sensitivity, we use the same 40 multiplier pairs and four starts at each $K = |\mathcal{W}_1| = |\mathcal{W}_2| \in \{2, 3, 4\}$. The cumulative pool starts with the binary pre- and post-SQP strategies. For $K = 3, 4$, we add the strategies returned by

the search, including its accepted SQP refinements, but do not separately add the pre-SQP endpoints. Smaller-alphabet strategies remain in the pool. Seed labels are shared, but random channels necessarily have different dimensions. The first case still improves by 0.00368 bit per symbol when size four is added. This measures the gain from the additional larger-alphabet search, not an isolated effect of cardinality, and motivates testing beyond binary auxiliaries. Independent enumeration of every retained controlled strategy gives errors below 7.9×10^{-16}.

F. A Target Correlated with the Side Information

To move beyond the common-mask XOR setting, keep $p_1 = 0.10$, $p_2 = 0.15$, and $Y \sim \mathrm{Bern}(1/2)$, but set $T = X_1 \wedge X_2$, where $\wedge$ is binary AND. The observations remain independent given Y, whereas

$$\mathbb{P}\{T = 1 \mid Y = 0\} = 0.015, \quad \mathbb{P}\{T = 1 \mid Y = 1\} = 0.765, \tag{49}$$

and $a = 0.39$. Thus the side information directly predicts the target, rather than only identifying a common observation mask.

We newly search 60 multiplier pairs with 12 logarithmically spaced $\lambda \in [0.4, 40]$, the five ratios above, four starts, and binary auxiliaries. The seed starts at 20761009. Retaining pre- and post-SQP points and completing decoder frontiers gives 256 strategies. Figure 5 shows the full achievable rate and fixed-rate sections, while Fig. 4 in the main text compares the rate sections with (36). The perception window $0 \le P \le 0.11$ covers the active range of the zero-message benchmark, and is a display choice rather than a change to the criterion.

With no messages, let $\theta_y = \mathbb{P}\{\hat{T} = 1 \mid Y = y\}$. Posterior sorting first allocates reconstruction ones to $Y = 1$, whose target posterior is larger. In the displayed range, the optimal marginal is $b = \min\{1/2, a + P\}$, with $\theta_0 = 0$ and $\theta_1 = 2b$. Hence the exact zero-message distortion is

$$D_{0,\wedge}(P) = a + (1 - 2 \cdot 0.765) \min\{1/2, a + P\}. \tag{50}$$

It decreases from 0.1833 at $P = 0$ to 0.125 at $P = 0.11$. Unlike the XOR example, increasing the reconstruction marginal above a can help. Both signs of the marginal multiplier are consequently meaningful in the search. Independent scalar enumeration and mixture replay agree within 3.4×10^{-16}, and the maximum budget excess is below 8.0×10^{-11}. The conditional program (36) gives 0.160326 bit per symbol at $(D, P) = (0.06, 0)$, whereas the saved pool gives 0.433598 bit per symbol.

The zero-message endpoint also illustrates marginal cancellation. The decoder that always outputs zero has $(D, b) = (0.39, 0)$, whereas the decoder that outputs Y has $(D, b) = (0.125, 0.5)$. Mixing them with weights 0.22 and 0.78 gives $b = 0.39$ and $D = 0.1833$, meeting $P = 0$. Averaging their individual perception losses would instead give $0.22(0.39) + 0.78(0.11) = 0.1716$. Thus storing only component perception losses would exclude this valid mixture, whereas the marginal constraints in (45) admit it.

G. Additional Three-Dimensional Views

Figure 6 retains the three XOR surfaces omitted from the five-page main text. Its two horizontal coordinates are budgets, not necessarily the distortion and perception attained with equality by the selected mixture. No smoothing procedure changes the stored numerical values. The vector PDF is used to avoid raster pixelation when enlarged.

H. Discussion and Extensions

Structured function codes can outperform unstructured descriptions in some distributed settings [6]. A tighter distributed converse than the cooperative relaxation would help distinguish incomplete channel search from limitations of the product construction.

For nonbinary targets, the mixture step can retain the full reconstruction pmf and express total variation using linear epigraph constraints. A vector of marginal multipliers then replaces the binary scalar ν. The posterior-sorting decoder rule is specific to binary Hamming loss and would require a more general decoder LP. These extensions, together with auxiliary-cardinality bounds, are natural next steps.

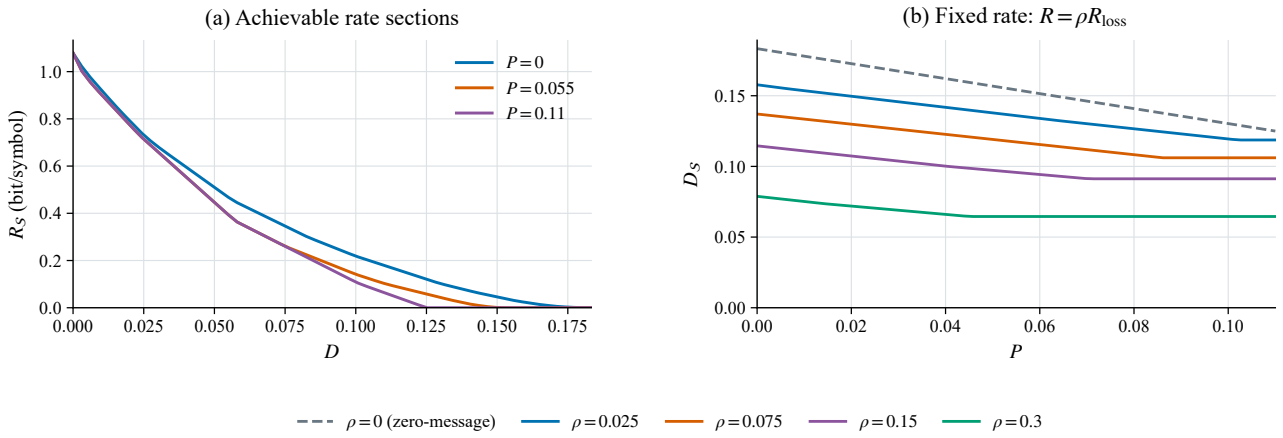


Fig. 5. Newly computed achievable tradeoffs for $T = X_1 \wedge X_2$, with $(p_1, p_2) = (0.10, 0.15)$ and $a = 0.39$. Left: rate versus distortion at three perception budgets. Right: fixed-rate DP sections with $R = \rho R_{\text{loss}}$. The dashed curve is (50), which uses decoder side information even with no messages.

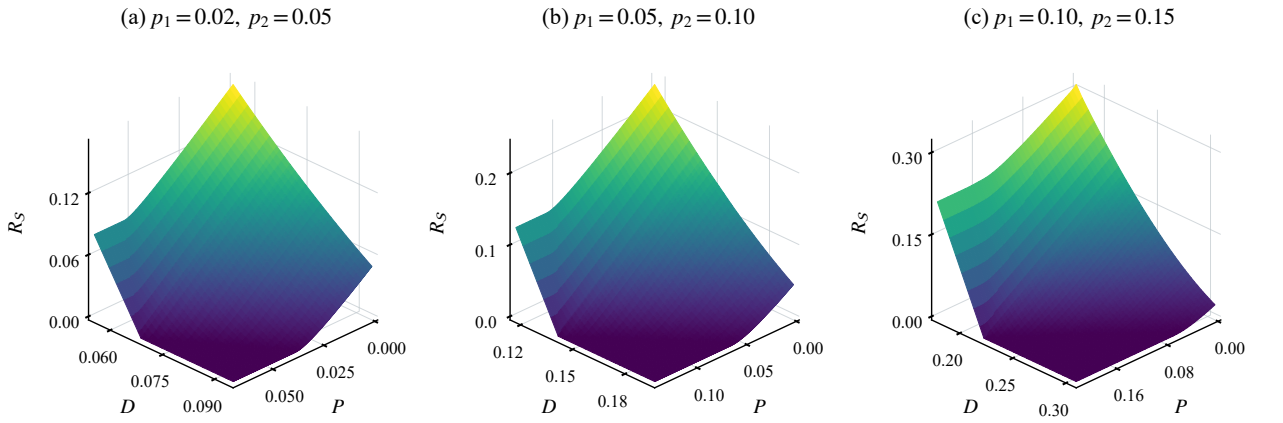


Fig. 6. Original vector-rendered XOR surfaces $R_S(D, P)$ over $0.8a \le D \le 1.4a$. Height and color show achievable sum rate. Faces join adjacent saved budget-grid values, without fitting new experimental points. These views supplement the upper–lower comparisons in Fig. 2.